Software Process Management

Lecture 9

Descriptive Process Models

- Descriptive process models are a mechanism of capturing the processes being pursued by an organization.

- We will be discussing:
- Typical goals of descriptive process modelling efforts
- A systematic approach for creating a descriptive process model
- Eliciting process information through interviews
- The management of risks associated with descriptive process modelling efforts.

Why Descriptive

- The downside of prescriptive models is that, being generic, they cannot exactly fit the particularities of a specific organization.

- In practice, no organization exactly follows a generic prescribed process, and no two organizations operate according to exactly the same process.

- The actual processes present in an organization, those that are used in practice to produce and deliver the organization's products and services, can usually be counted as one of its main assets.
- Normally, these processes encompass a good deal of the experience and knowledge that make an organization successful.
- For this reason, managing them appropriately can become a crucial activity for an organization's long-time survival.
- The discipline behind this type of process management is called descriptive process modelling.
- As its name implies, this discipline is concerned with producing an explicit and accurate representation of an organization's actual processes, for purposes such as documentation, dissemination, analysis, and improvement.

Goals of Descriptive Process Modelling.

- 1) Stable and Accurate Process Execution
- Software processes can be very complex, involving large numbers of interrelated activities.
- For every software development project, these activities must be carefully orchestrated, so that dependencies are respected and resources are available where and when they are needed.
- This complexity makes it very hard to guarantee that processes are always executed in the same way.
- Given the large number of individual activities and their complex interdependencies, there is a high risk that some activities are not planned or are skipped because of external pressure.
- Also, there is always the risk that people will change the process over time without even noticing it, maybe eliminating beneficial activities.

- A documented process helps to mitigate these risks.
- With a proper process description at hand, both project planners and project performers can make sure that they are performing all important and expected activities, and that they are taking into account all relevant interrelations between those activities.
- Also, a documented process makes it easier to introduce changes in a controlled fashion, which leads us to our next point: understanding the process.

2) Process Understanding

- Software development processes must change for a number of reasons. Sometimes, external factors force changes in the development process.
- Very often, however, the process is changed in order to introduce improvements. 
- In many organizations, this happens in an organic fashion, because people identify situations where work can be done in a better way.
- There is always the risk, however, that changes that are originally perceived as improvements end up having unintended, detrimental consequences, which may negatively affect the overall process performance.

- For example, tools are often introduced because they are expected to improve the process by automating parts of it.
- It may happen, however, that a particular tool is designed with a process in mind that is too different from that of the organization. In this case, the tool is likely to introduce inefficiencies that are not compensated by its potential advantages.

- Understanding the process is fundamental for managing this risk.
- An explicit process representation makes it much easier to assess the overall impact of process changes, thus allowing better management of process changes.

3) Process Propagation

- A common problem, present especially in large organizations, is that different development groups in the organization follow different processes for the same task.
- Sometimes, these process differences are related to the fact that different groups may have different needs.
- However, it often happens that they work differently simply because it is difficult for a group to exactly tell what other groups are doing.

- By explicitly describing the processes followed by various organizational units, it is easier to achieve a unified process.
- Differences between groups can still exist, but they are the result of conscious decisions based on analysing each group's particular needs.
- A unified process is important because it allows for practices that are widely accepted as being beneficial, to be propagated to all units in a large organization.

4) Process Measurement

- As an organization reaches a higher degree of process maturity, it becomes increasingly important to measure process performance.
- Aspects that are commonly measured in a process are its productivity (amount of output produced vs. resources invested into producing it), its ability to deliver results on time, and its ability to detect and correct product defects as early as possible, among many others.
- The reasons for measuring vary widely from one organization to another, but often include the desire to improve planning and increase the quality of software products by reducing the number of defects present in them when they are delivered.

- Measuring a process involves acquiring quantitative information about the actual activities, products, and resources involved.
- This is a hard task requiring high levels of discipline in order to yield satisfactory results, and which therefore must be carefully planned and executed.
- A proper process description greatly facilitates such planning by making it easier to identify appropriate measurement objects and to specify the procedures necessary to measure them.
- In addition, it provides the basis for comparing two processes with each other: It is only feasible to compare, for instance, the “duration of code development” if it is known which activities are contained: only coding, or also unit/integration testing?

5) Process Administration

- In the long term, one of the main reasons for explicitly describing an organization's development processes is to be able to define goals for the process and work systematically on achieving them.



- Doing this involves at least some of the descriptive process modelling goals explained earlier, such as process understanding and process measurement.

- Long-term goals for a process are often related to improvement (e.g., increase product quality, reduce time dedicated to product rework, etc.), but may be related to other process aspects.

- For example, many companies are forced by legal regulations, or, simply, by the critical nature of their products, to comply with existing norms and regulations regarding the way their products are produced and validated.
- For these organizations, one important reason for modelling their processes explicitly is to guarantee that they comply with whatever norms and regulations are considered relevant.
- Without a proper process description, this can be very difficult or even impossible to achieve.

6) Process Automation

- It happens quite often that once a process has been properly described and stabilized, opportunities for supporting parts of it with automated tools become evident.



- Analysing a detailed process description is probably one of the best ways to determine the exact requirements of new process-supporting tools.

- One area where processes can be supported in a particularly effective way is information flow.

- Since process models often contain detailed descriptions of the product and control flows in a process, a tool can effectively support process participants by giving them access to the pieces of information that are relevant for their activities and by providing appropriate mechanisms for storing new work products and informing other process performers about their availability and status.
- This type of support is especially useful in modern development environments, where developers are frequently distributed geographically, often across several time zones.

Creating a Descriptive Process Model

- Process engineers working on describing an organization's processes are confronted with two main challenges.
- The first of these challenges is the so-called process elicitation, namely, collecting information about how the process is actually performed.
- This can be done by working together with the actual process performers, by observing their work, or by analyzing the results of their work, such as the documents produced during past development projects.
- Through interviews, direct observation of people's work, and analysis of documents and final products, among other techniques, process engineers gain a detailed understanding of the process.

- The second challenge process engineers face is storing the process knowledge for future use.
- The process-related information must be expressed in a clear and unambiguous way that is amenable to the goals of the modelling effort (e.g., process measurement, improvement, analysis, etc.).
- A number of notations and modelling tools are available to facilitate this task.
- Notations allow expressing processes using a predefined, standardized set of concepts.
- They can be formal or graphical in nature, with graphical notations being preferred in recent years because they facilitate communication with process stakeholders.
- Tools, in turn, help process engineers to deal with the complexity of process models, which, in practice, may easily grow to contain hundreds or even thousands of individual elements.

8 Step process

- The approach consists of two major phases: a set-up phase configuring the modelling approach for the organization, the modelling goals and context; and an execution phase performing the actual modelling.
- The set-up phase consists of these four steps:
- 1. State objectives and scope
- 2. Select or develop a process modeling scheme
- 3. Select (a set of) process modeling formalisms
- 4. Select or tailor tools

8 Step process

- The execution phase consists of these four steps:
- 5. Elicitation
- 6. Create the process model
- 7. Analyse the process model
- 8. Analyse the process



- In theory, these steps should be repeated for every process modelling effort in order to achieve optimal results.
- In practice, however, the process modelling scheme, formalisms, and tools are likely to be rather stable across different modelling efforts, because in a real-world setting, it is usually not feasible to frequently switch them due to cost reasons.

1. State Objectives and Scope

- In order to properly plan and later manage the process modelling effort, it is thus important to state its goals clearly before any modelling has started.
- The scope of the effort must also be clearly stated.
- This includes specifying 
- – Who will use the process model
- – What these people expect of the model
- – Which processes should be covered
- – Which processes are explicitly excluded from the model

2. Select or Develop a Process Modeling Schema

- The second step of descriptive process modeling consists of identifying the set of concepts that will be used to describe processes.
- it is possible to identify a number of basic concepts that are important for describing processes in the organization:
- – Activity
- – Work product
- – Role
- – Product flow between activities and work products, i.e., which products are inputs or outputs to which activities
- – Assignment of roles to activities, i.e., which roles are responsible for performing which activities

- Usually, a set of concepts such as this used to model processes is called a process schema.
- Although identifying an appropriate set of modeling concepts may appear to be quite simple from this small example, it is, in practice, a very difficult task, and one that deserves very careful consideration given its importance for the success of the subsequent modeling steps.
- As a minimum, the following requirements should be taken into account while choosing a process schema:

- The concepts must cover those aspects of a process that are relevant for the current modeling effort.
- The concepts must allow for an adequate level of detail and formalism.
- The concepts must facilitate modeling.
- Modeling results must be accessible for their intended audience.
- Designing a set of concepts “from scratch” that fulfills all of these requirements can be a difficult task even for experienced process engineers.
- For this reason, it is strongly advisable to use an existing schema whenever possible.

3 Select (a Set of) Process Modeling Formalisms

- Process modeling not only requires an appropriate set of concepts but a concrete notation in which these concepts can be expressed.
- A variety of modeling notations have been used over time, with significant differences in formalism and expressive power.
- Process notations can be characterized along a number of dimensions:
- Graphical vs. textual. Some notations rely mainly on text, whereas others are mainly graphical. Some combine these aspects to various extents.

- Formal vs. informal. Some notations rely on mathematically defined elements, which can only be combined according to strict rules, whereas others rely on natural language text and loosely defined graphical elements.
- Formal elements are often combined with informal ones to achieve a richer description that is easier to understand.
- Fine grained vs. coarse grained. Some notations are intended to describe processes at a macro level, whereas others provide support for producing very detailed descriptions.

- Predetermined vs. extensible. Some notations are intended to be used “as-is,” with no modifications whatsoever, whereas others allow for extensibility. Extensibility, in turn, may range from very basic, allowing only for simple tailoring that is useful only in certain common cases, to very advanced, allowing for tailoring that can handle almost any practical situation imaginable.

- Taking these dimensions into account, it is possible to choose a notation that is appropriate for the needs of a particular modeling effort.

- Similar to the identification of modeling concepts discussed in the previous step, selecting an appropriate notation is a demanding task. Indeed, a set of requirements analogue to the one presented for the previous step applies here as well:
- – The notation must cover the concepts selected in the previous step.
- – The notation must allow for an adequate level of detail and formalism.
- – The notation must facilitate modeling.
- – The modeling results (as represented in the chosen modeling notation) must be accessible to their intended audience.

- Given the difficulty of finding appropriate modeling notations, this step is often interwoven in practice with the previous one, that is, the detailed set of modeling concepts and the notation are selected simultaneously.
- Still, thinking of both steps as conceptually separate remains useful.

4 Select or Tailor Tools

- Tool support is usually necessary to effectively model complex processes.
- Among other things, in a descriptive modeling effort, tools are useful for:
- - Supporting the use of the chosen modeling notation.
- - Storing large models or model collections in a convenient and safe way.
- - Producing alternative representations of models for different stakeholders.

- One important aspect about tools is that they do not have to be specialized process modeling tools in order to be adequate for a particular modeling effort.
- In many cases, standard office tools (word processor, spreadsheet) and/or generic diagramming tools can be effectively used to model processes, as long as the chosen notation is simple enough for them and the models are small and need only little maintenance.